Ulrich Aïvodji

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Main Research Interests

— Machine learning : privacy, fairness, explainability, interpretability
— Data privacy : privacy-enhancing technologies, location-based services
— Optimization : combinatorial optimization, transportation

Professional Experiences

Université du Québec à Montréal — **Montreal, Canada**
Postdoctoral researcher — *March 2018–Present*
- Advisor : Sébastien Gambs
- Privacy, Interpretability, and Fairness in Machine Learning

Université du Québec à Montréal — **Montreal, Canada**
Visiting researcher — *July–August 2016*
- Advisors : Sébastien Gambs, Jean-Marc Robert
- Secure multi-party computation for linear programming

Laboratoire d'Analyse et d'Architecture des Systèmes (LAAS-CNRS) — **Toulouse, France**
Ph.D. Candidate — *Oct. 2014–Jan. 2018*
- Advisors : Marie-José Huguet, Marc-Olivier Killijian
- Privacy-preserving algorithms for route synchronization and ride-matching, for ridesharing

Laboratoire d'Analyse et d'Architecture des Systèmes (LAAS-CNRS) — **Toulouse, France**
Research Intern — *March–August 2014*
- Distributed and privacy-preserving algorithms for route synchronization

Dial Technologies — **Casablanca, Morocco**
Software Engineering Intern — *July–August 2013*
- Design and implementation of a Backend as a Service (BaaS) for mobile applications

Multi Information Technologies — **Marrakesh, Morocco**
Software Engineering Intern — *June–August 2012*
- Design and implementation of an Enterprise Resource Planning (ERP) software

Comtel Technologies — **Cotonou, Benin**
Software Engineering Intern — *June–July 2010*
- Initiation to Network Administration

Education

Université Paul Sabatier — **Toulouse, France**
Ph.D. in Computer Science — *2014–2018*
- Dissertation : Privacy-enhancing technologies for ridesharing
- Advisors : Marie-José Huguet, Marc-Olivier Killijian
- Rapporteurs : Josep Domingo-Ferrer, Dominique Feillet
- Committee members : Cyril Briand, Françoise Fessant, Bertrand Le Cun, Benjamin Nguyen

Ecole Nationale des Sciences Appliquées — **Khouribga, Morocco**
Engineer's degree in Software Engineering — *2009–2014*
Valedictorian, GPA : 17.10/20

Certificates and Accreditations

French Ministry of Research and Education (Conseil National des Universités) — **2020**
Qualification for assistant professor position
- Scientific section 27
- Qualification number : MCF-27-20227342699

Stanford University — **2017**
Machine learning class (Certificate link)

University of Maryland, College Park — **2015**
Cryptography class (Certificate link)

University of Maryland, College Park — **2015**
Usable Security class (Certificate link)

Software Skills

Programming: Python, C++, Java, C#, Scala, R, Julia

Machine learning & Optimization: Scikit-learn, Pytorch, Tensorflow, Flux, Or-tools, CPLEX, JuMP

Web and mobile development: Django, Meteor, Android, Flutter

Awards

Travel Award: 36th International Conference on Machine Learning *2019*

Travel Award: 11th ACM Conference on Security and Privacy in Wireless and Mobile Networks *2018*

Ph.D. Scholarship: École Doctorale MITT ($68,000$ euros, ranked 1st out of 23 candidates) *2014-2017*

Visiting scholar grant: Université Fédérale Toulouse Midi-Pyrénées *2016*

AMCI Scholarship: Agence Marocaine de Coopération Internationale ($45,000$ dirhams) *2009-2014*

Medal for exceptional academic achievements: Université Hassan 1er *2014*

Innov'IT (2nd place): AUSIM Maroc ($40,000$ dirhams) *2013*

Startup Weekend (1st place): Startup Weekend Casablanca *2013*

Medal for exceptional academic achievements: Université Hassan 1er *2013*

Publications

Journals and Conference proceedings

Local Data Debiasing for Fairness Based on Generative Adversarial Training. *Ulrich Aïvodji, François Bidet, Sébastien Gambs, Rosin C. Ngueveu, and Alain Tapp.* Algorithms, 14 (3), 87, 2021

Learning-based Incast Performance Inference in Software-Defined Data Centers. *Kokouvi Benoit Nougnanke, Yann Labit, Marc Bruyere, Simone Ferlin, and Ulrich Aïvodji.* Proceedings of the 24th Conference on Innovation in Clouds, Internet and Networks (ICIN), Virtual conference, 2021. **Best Paper Award**

Privacy in trajectory micro-data publishing : a survey. *Marco Fiore, Panagiota Katsikouli, Elli Zavou, Mathieu Cunche, Françoise Fessant, Dominique Le Hello, Ulrich Aïvodj, Baptiste Olivier, Tony Quertier, and Razvan Stanica.* Transactions on Data Privacy 13 (2), 91 - 149, 2020

Fairwashing : the risk of rationalization. *Ulrich Aïvodji, Hiromi Arai, Olivier Fortineau, Sébastien Gambs, Satoshi Hara, and Alain Tapp.* Proceedings of the 36th International Conference on Machine Learning (ICML), Long Beach, CA, 2019

SRide : a privacy-preserving ridesharing system. *Ulrich Aïvodji, Kévin Huguenin, Marie-José Huguet, and Marc-Olivier Killijian.* Proceedings of the 11th ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec), Stockholm, Sweden,2018

Meeting points in ridesharing : a privacy-preserving approach. *Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Marc-Olivier Killijian.* Transportation Research Part C : Emerging Technologies, 72, 239-253, 2016

Book chapters

Les enjeux éhiques de l'apprentissage machine. *Ulrich Aïvodji and Sébastien Gambs.* 13 défis de la cybersécurité. Gildas Avoine and Marc-Olivier Killijian (Dir.), CNRS Éditions 2020.

Workshops

GAMIN : an adversarial approach to black-box model inversion. *Ulrich Aïvodji, Sébastien Gambs, and Timon Ther.* AAAI Workshop on Privacy-Preserving Artificial Intelligence, New York, 2020

IOTFLA : a secured and privacy-preserving smart home architecture implementing federated learning. *Ulrich Aïvodji, Sébastien Gambs, and Alexandre Martin.* IEEE Workshop on the Internet of Safe Things, San Francisco, CA, 2019

SRide : a privacy-preserving ridesharing method. *Ulrich Aïvodji, Marie-José Huguet, and Marc-Olivier Killijian.* 19ème congrès annuel de société Française de Recherche Opérationnelle et d'Aide à la Décision, Lorient, France, 2018

Privacy-preserving matching for ridesharing. *Ulrich Aïvodji, Kévin Huguenin, Marie-José Huguet, and Marc-Olivier Killijian.* 8e édition de l'Atelier sur la Protection de la Vie Privée, Autrans, France, 2017

Covoiturage respectueux de la vie privée. *Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Marc-Olivier Killijian.* 17ème congrès annuel de société Française de Recherche Opérationnelle et d'Aide à la Décision, Compiègne, France, 2016

Privacy-preserving carpooling. *Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Marc-Olivier Killijian.* 6th International Workshop on Freight Transportation and Logistics, Ajaccio, France, 2015

Privacy-preserving route synchronization for ridesharing. *Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Marc-Olivier Killijian.* 6e édition de l'Atelier sur la Protection de la Vie Privée, Mosnes, France, 2016

Pre-prints

Model extraction from counterfactual explanations. *Ulrich Aïvodji, Alexandre Bolot, and Sébastien Gambs.* [arXiv:2009.01884]

Hiccups on the road to accountable machine learning : the risk of ethics washing. *Ulrich Aïvodji, Hiromi Arai, Céline Castets-Renard, Sébastien Gambs, and Dominic Martin.*

Learning fair rule lists. *Ulrich Aïvodji, Julien Ferry, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala.* [arXiv:1909.03977]

Teaching

École Nationale de l'Aviation Civile (19.5 hours)

Lecture: Computer Security *Fall 2017*

Institut National des Sciences Appliquées de Toulouse (123.5 hours)

Lab sessions: Algorithms, Complexity, Graph Theory, Database, Data Analysis, Unix *2015-2017*

Talks

Guest Lectures

Privacy and security risks in machine learning: IFT6758, Université de Montréal *2020*

Privacy and security risks in maching learning: INF8750, Université du Québec à Montréal *2020*

Privacy and Transparency in Machine Learning: IFT6758, Université de Montréal *2019*

Location privacy: INM6000, Université du Québec à Montréal *2018*

Invited Talk

Fairwashing and model extraction : Two challenges for XAI: Carrefour DEEL #4, Montreal (Canada) *2021*

Learning Fair Rule Lists: RE·WORK Responsible AI Summit 2019, Montreal (Canada) *2019*

Fairwashing : the risk of rationalization: SÉSÀM 2019, Montreal (Canada) *2019*

Transparency and privacy in the era of big-data: Toulouse Data Science Meetup, Toulouse (France) *2017*

Privacy Enhancing Technologies for ridesharing: Université du Québec à Montréal, Montreal (Canada) *2016*

Privacy-preserving two-points synchronization for ridesharing: IRISA, Rennes (France) *2015*

Academic Services

Program committee

APVP: Atelier sur la Protection de la Vie Privée *2020*

Reviewer

International Conference on Machine Learning : *2021*

IEEE Transactions on Knowledge and Data Engineering: *2020*

IEEE Transactions on Computers: *2020*

IEEE Transactions on Mobile Computing: *2020*

Privacy Enhancing Technologies Symposium: *2020*

IEEE Software: *2019*

Network Management: *2018*

Review assistance

IJCAI-ECAI: *2018*

Open source projects and popular science

Volunteering

Mentor at UpstartED: coaching college students on privacy-preserving ML projects *Since Feb. 2020*

Software

Co-author of FairCORELS: open-source library for learning fair rule lists *2019*

Co-author of LaundryML: open-source library for analyzing ML explanations *2019*

Popular Science

Fête de la science: Demo of a privacy-preserving ride-sharing system developed during my Ph.D. *2016*

Ma thèse en 180 secondes: Scientific vulgarization challenge of presenting Ph.D. research in 180 seconds *2015*

Open Source Days: Co-design the website of the event. Demo of an AIML-based chatbot *2012*